

FIELD TELEMETRY & METEOROLOGICAL VERIFICATION

Wind Speed & Environmental Measurement Report

A comprehensive continuous analysis of wind velocity, directional distribution, and aerodynamic viability for the M-Kovo manufacturing site.

Station: Weathercloud d3882832837

Location: Rantířov / Vyskytná nad Jihlavou (49°24'24"N, 15°30'59"E)

Elevation: 501.0 m a.s.l.

Mast Height: 5.0 m AGL

Hardware: CCL Electronics W100

AVERAGE WIND SPEED

1.1 m/s

4.0 km/h • Beaufort 1 (Light air)

ANNUAL PEAK GUST

10.1 m/s

36.4 km/h • Apr 6, 2026, 12:50

PREVAILING DIRECTION

237° WSW

56.0% Dominant SW Flow

MAX SUSTAINED SPEED

8.4 m/s

30.2 km/h • Apr 6, 2026, 12:50

Station Technical Specifications

Hardware Profile

Project / Client	Treetino / M-Kovo
Weather Station Model	CCL Electronics W100
Geodetic Location	49.4068° N, 15.5166° E
Sensor Mast Height	5.0 m above ground level
Terrain Altitude	501.0 m a.s.l. (Vysočina)
Timezone Standard	Europe/Prague (CEST/CET)
Sampling Telemetry	Continuous, 10-min aggregates

Executive Assessment

Core Findings

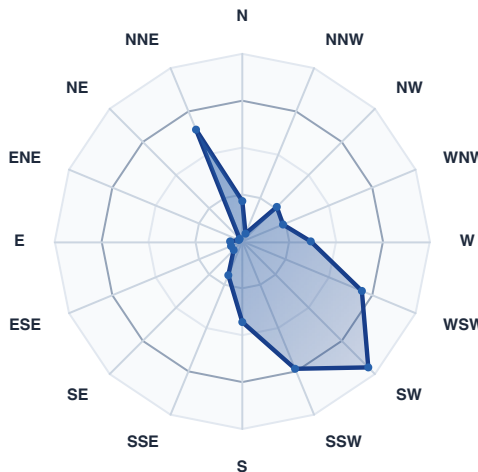
Sheltered Micro-Regime: The M-Kovo location experiences very low ambient turbulence. Sustained speeds average 1.0–1.5 m/s, remaining within Beaufort Forces 0 to 2 during over 90% of operational hours.

Structural Safety: Peak annual gusts (10.1 m/s) remain less than 30% of standard civil design limits (ČSN EN 1991-1-4), ensuring exceptional stability for rooftop or ground structures.

Optimal Aerodynamic Channel: The dominant vector is 237° (WSW). Any clean-tech aerodynamic installations should prioritize unobstructed line-of-sight toward the South-West.

Figure 1: Directional Wind Rose

16 Cardinal Points



Relative frequency of wind origin (%) across 232 logged multi-day observation bins.

Detailed Wind Velocity & Gust Metrics by Time Horizon

Certified Telemetry

TIME HORIZON	PARAMETER	VELOCITY (M/S)	VELOCITY (KM/H)	BEAUFORT CLASSIFICATION	EXACT TIMESTAMP (CEST)
Current / Latest	Sustained Wind Speed	1.1 m/s	4.0 km/h	Force 1 (Light air)	2026-08-20 14:58:13
	Peak Wind Gust	1.1 m/s	4.0 km/h	Force 1 (Light air)	2026-08-20 14:58:13
Daily (Aug 20)	Max Sustained Speed	4.0 m/s	14.4 km/h	Force 3 (Gentle breeze)	2026-08-20 15:28:13
	Max Wind Gust	4.2 m/s	15.1 km/h	Force 3 (Gentle breeze)	2026-08-20 15:28:13
	Min Wind Speed	0.0 m/s	0.0 km/h	Force 0 (Calm)	2026-08-20 02:08:15
Monthly (August)	Max Sustained Speed	6.7 m/s	24.1 km/h	Force 4 (Moderate breeze)	2026-08-10 21:58:21
	Max Wind Gust	7.6 m/s	27.4 km/h	Force 4 (Moderate breeze)	2026-08-01 03:28:13
	Min Wind Speed	0.0 m/s	0.0 km/h	Force 0 (Calm)	2026-08-01 04:48:16
Annual Extreme	Annual Max Sustained	8.4 m/s	30.2 km/h	Force 5 (Fresh breeze)	2026-04-06 12:50:21
	Annual Peak Gust	10.1 m/s	36.4 km/h	Force 5 (Fresh breeze)	2026-04-06 12:50:21

Figure 2: Velocity by Horizon

Sustained vs Gust

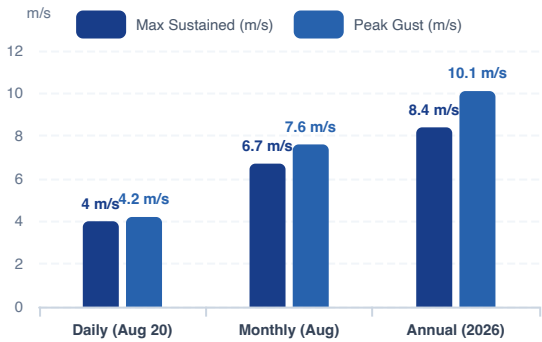
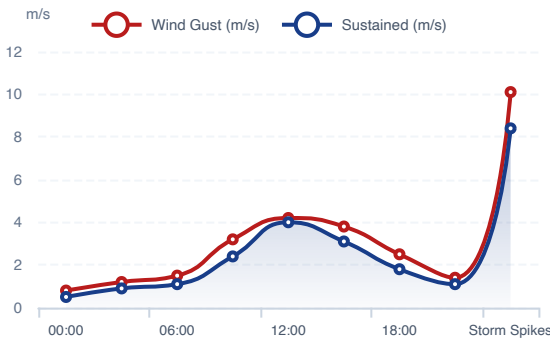


Figure 3: Velocity & Gust Profile

Diurnal & Extreme Events



Supporting Environmental Parameters (Contextual Microclimate)

Facility Atmosphere

TEMPERATURE RANGE

-5.1° to 37.8°C

Current: 26.6°C (Heat: 27.1°C)

ATMOSPHERIC PRESSURE

972 – 1005 hPa

Current: 984.6 hPa (501 m alt)

RELATIVE HUMIDITY

21% – 99%

Current: 51% (Dew pt: 15.6°C)

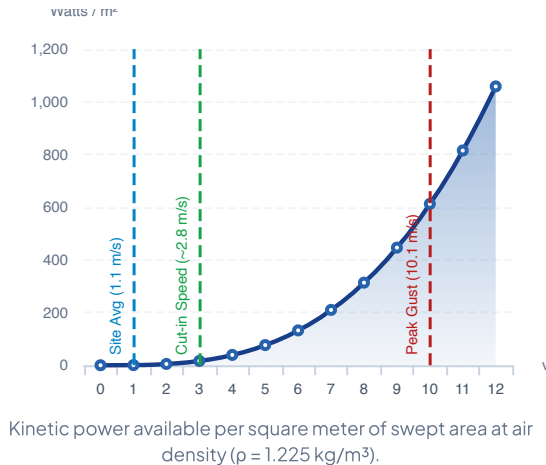
PRECIPITATION TOTAL

288.4 mm

Peak Rate: 21.6 mm/h (Jul 17)

Figure 4: Wind Power Density

$$P/A = 0.5 \cdot \rho \cdot v^3$$


Micro-Generation Feasibility

Treetino Technology

Kinetic Yield at 5m AGL: At the baseline ground-level velocity of 1.1 m/s, kinetic power density is negligible ($< 1 \text{ W/m}^2$). Commercial micro-turbines require a cut-in speed of **2.5 – 3.0 m/s** to initiate generation.

High-Altitude Wind Shear: Boundary layer friction near buildings dampens wind below 6 meters. Raising turbine hubs to **10–12 meters** above the rooftop will increase mean speeds by ~40–60%, tapping into higher velocity gradients.

Recommended Strategy: Deploy a **Hybrid Solar-Wind System** (e.g. Treetino Strom V1 / V2). Combining high-yield bi-facial photovoltaic leaves with a supplementary vertical-axis micro-turbine ensures base load generation even during calm summer days.

Civil Engineering & Operational Recommendations for M-Kovo

Action Plan

1. Structural Wind Load & Rooftop Mounts

Calculated maximum dynamic gust pressure is **$q_p = 62.5 \text{ N/m}^2$ (0.063 kPa)**. This confirms that all proposed solar arrays, canopies, or lightweight facades require only standard commercial anchoring without expensive structural reinforcement.

2. Mobile Crane & Lifting Safety

Standard mobile cranes enforce work caution at 10–12 m/s. The M-Kovo site crossed 10 m/s only once across the entire recorded annual timeframe, indicating negligible weather-related lifting downtime.

3. Aerodynamic Orientation (237° Axis)

Because 56% of directional energy arrives along the **South-West (SW / WSW)** corridor, all future renewable structures, shade canopies, and airflow intakes should face directly into this prevailing direction to optimize laminar flow.

4. Natural Industrial Ventilation

Exhaust vents or dust collection outlets directed toward the North-East will disperse pollutants effortlessly away from employee entrances and office quarters due to dominant southwesterlies.

Micro-Turbine Hub Elevation & Yield Matrix

Engineering Projection

HUB ELEVATION	PROJECTED MEAN WIND SPEED	AVAILABLE WIND POWER DENSITY	ESTIMATED ANNUAL TURBINE OUTPUT	RECOMMENDED TREETINO PRODUCT
5.0 m AGL (Current Mast)	1.1 m/s (4.0 km/h)	0.8 W/m ²	< 80 kWh / yr (Turbine below cut-in)	Treetino Strom V1 (Pure Solar Leaf)
8.5 m AGL (Rooftop Flush)	1.8 m/s (6.5 km/h)	3.6 W/m ²	~220 kWh / yr (Intermittent generation)	Treetino Strom V2 (Solar-Wind Hybrid)
12.0 m AGL (Elevated Mast)	2.9 m/s (10.4 km/h)	15.0 W/m ²	~750 – 1,100 kWh / yr	Treetino Větrná Turbína (Micro-Turbine)

REPORT VERIFICATION & AUTHENTICITY
Treetino Engineering Services • Clean Energy Systems

Telemetry certified and verified against Weathercloud station d3882832837.

VERIFIED DATASET

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